

Ruoxiao (Roxanna) Zhang

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SUMMARY OF QUALIFICATIONS

- Honours Applied Mathematics (Scientific Computing and Scientific Machine Learning) co-op student at the University of Waterloo: **88.4% cumulative average**, Term Distinction every term, President's Scholarship of Distinction.
- Frontend engineering intern at Xiaomi**, delivering production features on a React/TypeScript micro-frontend cloud platform and driving team-wide test and CI reliability across 298 repositories.
- AI-augmented engineer with a USACO Gold foundation: builds and evaluates LLM agent tooling (Claude Code skills, RAG), and launched two data products: a Monte Carlo football forecaster and a luxury market tracker.

EXPERIENCE

Xiaomi Technology

Sept. 2026 – Present

Frontend Engineering Intern, Cloud Frontend Team

Wuhan, China

- Delivered production features** for Xiaomi's internal cloud platform, a React/TypeScript micro-frontend system, spanning data export, permission workflows, deep linking and mobile-responsive layouts
- Drove a team-wide CI reliability effort across 298 repositories**, restoring 24 broken pipelines via GitLab API automation and taking a core project from 5 straight failed builds to a 100% pass rate
- Architected Playwright end-to-end testing infrastructure**, raising E2E coverage from 0% to 71% and nearly doubling another app's (24.6% → 42.2%), while fixing 30+ false-positive tests
- Built AI-augmented developer tooling**: Claude Code agent skills with automated evals, an AI page-generation system scaled to a 151-component design library, and a RAG vs. memory-retrieval benchmark

PROJECTS

UCL Simulator – Champions League Forecaster | *Python, JavaScript, Claude Code*

Oct. 2026 – Present

- Built a full-stack forecasting app for the 2026/27 Champions League: a Python Monte Carlo engine that plays out the 144-match league phase and knockout bracket 50,000 times on an Elo and Poisson goal model
- Shipped a bilingual interactive dashboard (title-odds trends, round-by-round heatmaps, matchup score grids) whose client-side JavaScript engine re-runs 10,000 simulations in-browser on user-edited ratings
- Automated a fault-tolerant data pipeline across ESPN, ClubElo and Polymarket APIs (retries, graceful fallback, result preservation), refreshed three times per matchweek

Relay Resale Ledger | *Python, JavaScript, HTML/CSS, SVG, Claude Code*

Sept. 2026 – Present

- Designed and launched a bilingual (English/Chinese) market-intelligence web app tracking resale value for **98 luxury assets across 26 brands**, with interactive price-trend visualizations and a valuation engine
- Engineered an automated, fault-tolerant Python data pipeline (robots.txt-compliant scraping, anomaly detection, graceful fallback, full price history), refreshed weekly by a scheduled cloud agent

Data Processing & Statistical Computing Scripts | *Coursework Project, ECON 102*

2026

- Developed automated data processing scripts in R to parse and clean real-world economic datasets, implementing programmatic filtering and time-series decomposition algorithms
- Engineered robust data pipelines to extract business-cycle trends via MA(5) smoothing, generating automated statistical visualizations for cross-country growth analysis

FitFuel – Health & Fitness App | *Figma, UI/UX Design*

Jan. – Apr. 2024

- Designed 12 high-fidelity interactive Figma screens for a nutrition-tracking app, using survey-based user research to shape onboarding and macronutrient charts (Dartmouth–Chongqing University HCI course)

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Mathematics, Applied Mathematics with Scientific Computing and Scientific Machine Learning, Honours, Co-op

Sept. 2025 – Present

- Cumulative Average:** 88.4% (1A 85.6% | 1B 90.0% | 2A 89.6%); Term Distinction every term
- Relevant Coursework:** Intro CS 2 (90%), Designing Functional Programs, Probability (91%), Linear Algebra 2 (90%), Calculus 3 (89%), Algebra (92%)

Chongqing Nankai Secondary School

Chongqing, China

High School Diploma

Sept. 2022 – July 2025

LEADERSHIP & ACTIVITIES

International Space Settlement Design Competition

2023 – 2024

Human Factors Department Lead

Florida, USA / China

- Led a 10-person department within a 48-member design “company” to **three consecutive championships** (China Finals, Asian Regional Finals, International Finals in Florida)
- Owned residential planning and human-systems requirements under 24–42 hour design sprints, computing land-area and population-capacity requirements for a Mars settlement in Fusion 360

Official Student Ambassador

Apr. 2026 – Present

University of Waterloo, Undergraduate Admissions

Waterloo, ON

- Advise prospective and incoming students and families on courses, admissions and the move to university, and support online engagement for the incoming cohort

SKILLS & PUBLICATIONS

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, R

Frontend: React, Ant Design, Module Federation (micro-frontends), responsive HTML/CSS, SVG data visualization

Testing & DevOps: Playwright (E2E, visual regression), unit testing, GitLab CI/CD, Git, Node.js

AI & Design Tools: Claude Code, agent skills and evals, RAG (RAGFlow), Figma

Publication: Sole Author, “*The Impact of AI Represented by ChatGPT on UI Design*”, Proc. 2nd Intl. Conf. on Software Engineering and Machine Learning, 2024

AWARDS & COMPETITIONS

- **USA Computing Olympiad (USACO)** – Gold Division (2024)
- **AMC 12** – Top 5% (2023); **Euclid Mathematics Contest** – Top 25% (2023)
- **University of Waterloo President’s Scholarship of Distinction** (2026)